

Final exam

This is a *90 minutes* open-book exam. Please read the following instructions carefully.

- The exam consists of 7 pages including this cover page. Please make sure you have all the pages before you start.
- You are allowed to use your cheat sheet (one double-sided A4 paper).
- You are *not* allowed to use any electronic devices (*e.g.*, mobile phones, tablets, laptops) during the exam.
- Please write your answers in *English* using an *indelible pen* in the provided *answer sheets*, and make sure to write your name and matriculation number on *each* answer sheet before the exam starts.
- Please write your solutions following the problem numbering on the answer sheets. If you need more space, please ask for additional answer sheets and clearly indicate where the continuation is.
- Please submit *only* your answer sheets after the exam, and you may keep the exam paper (for, *e.g.*, the exam review). This means that you can use the back (or in theory, any blank part) of the exam paper for rough work, but we will not grade anything written there.
- Please write clearly and structure your answers well. We require complete clarity as to what you are arguing or doing in your solutions. We may not be able to give you full credit for answers that are hard to read or follow.

1. *Multiple choice (20 points).* Please mark each listed statement as true (✓) or false (✗). For each answer that is correctly marked as either true or false you will get 1 point, resulting in 4 points per question.

(a) Let $Y \in \{1, \dots, K\}$ be a class variable and $X = (X_1, \dots, X_d)$ a feature vector. A maximum a posteriori Bayesian classifier predicts $\hat{y}(x) = \operatorname{argmax}_y \mathbf{P}(Y = y \mid X = x)$. Which of the following statements are true?

- ☐ Changing the class prior $\mathbf{P}(Y = y)$ can change $\hat{y}(x)$ even if every class-conditional density $p(x \mid y)$ remains fixed.
- ☐ The naive Bayes assumption is the marginal factorization

$$p(x_1, \dots, x_d) = \prod_{i=1}^d p(x_i).$$

- ☐ Gaussian naive Bayes models can be directly applied to continuous features without discretization.
 - ☐ In a tree-augmented naive Bayes classifier, the feature subgraph may contain an arbitrary number of feature parents for each X_i , so that $p(x \mid y) = \prod_{i=1}^d p(x_i \mid y, \mathbf{pa}(X_i))$ without any restriction on $|\mathbf{pa}(X_i)|$.
- (b) Consider a finite-state Markov chain with transition matrix P , and a hidden Markov model with hidden states $X_{1:T}$ and observations $Y_{1:T} = y_{1:T}$. Which of the following statements are true?

- ☐ The forward algorithm computes $p(y_{1:T}) = \sum_{x_{1:T}} p(x_{1:T}, y_{1:T})$ recursively.
- ☐ In general,

$$\operatorname{argmax}_{x_{1:T}} p(x_{1:T} \mid y_{1:T})$$

need not equal the sequence formed by $\operatorname{argmax}_{x_t} p(x_t \mid y_{1:T})$ for each t .

- ☐ Baum-Welch parameter learning is an instance of the expectation-maximization algorithm.
 - ☐ For every transition matrix P , there exists a unique probability vector π satisfying the stationary distribution condition $P^T \pi = \pi$.
- (c) Let $G = (V, E)$ be a Bayesian network. Which of the following statements are true about inference and learning in G ?

- ☐ If $X_A \rightarrow X_C \leftarrow X_B$ is a collider, conditioning on X_C can make $X_A \not\perp\!\!\!\perp X_B \mid X_C$.
- ☐ Consider the variable elimination algorithm for the inference of a fixed query $p(X_Q \mid X_O = x_O)$, changing the variable elimination order can change the sizes of intermediate factors but not the exact result.
- ☐ If the underlying undirected graph of G contains at most one path between any two nodes, belief propagation computes exact node marginals.
- ☐ Consider the parameter learning in Bayesian networks. Laplace smoothing increases the number of zero entries in an estimate $\hat{p}(X_i \mid \mathbf{pa}(X_i))$.

- (d) Suppose we want to estimate $\mu = \mathbf{E} f(X)$ with respect to $X \sim p$ for a target density $p(x) = \tilde{p}(x)/Z$, where the normalizing constant Z is unknown. We draw independent samples $x^{(1)}, \dots, x^{(N)}$ from a proposal density q and use the weights $w^{(i)} = \tilde{p}(x^{(i)})/q(x^{(i)})$ and $\bar{w}^{(i)} = w^{(i)} / \sum_{j=1}^N w^{(j)}$. Which of the following statements are true?

- ☐ A valid importance sampling proposal must satisfy $q(x) \geq p(x)$ for every x , and $q(x) > 0$ whenever $p(x) > 0$.
- ☐ Replacing every weight $w^{(i)}$ by $cw^{(i)}$ for a constant $c > 0$ changes the value of the self-normalized estimator.
- ☐ For every finite N , the estimator

$$\hat{\mu}_N = \sum_{i=1}^N \bar{w}^{(i)} f(x^{(i)})$$

satisfies $\mathbf{E} \hat{\mu}_N = \mu$.

- ☐ If $q = p$, then $\bar{w}^{(i)} = 1/N$ for all i , and $\hat{\mu}_N$ reduces to the ordinary sample average.
- (e) Consider a finite discounted Markov decision process $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ with $0 \leq \gamma < 1$ and a policy $\pi(a \mid s)$. Which of the following statements are true?
- ☐ $V^\pi(s) = \mathbf{E}_\pi(G_0 \mid S_0 = s)$, where $G_0 = \sum_{t=0}^{\infty} \gamma^t R_t$.
 - ☐ A Bellman policy evaluation update for $V^\pi(s)$ takes a maximum over $a \in \mathcal{A}$.
 - ☐ Value iteration converges to the unique fixed point V^* of the Bellman optimality operator.
 - ☐ A maximum entropy control objective contains an expected return term and a term proportional to $\mathcal{H}(\pi(\cdot \mid s))$.

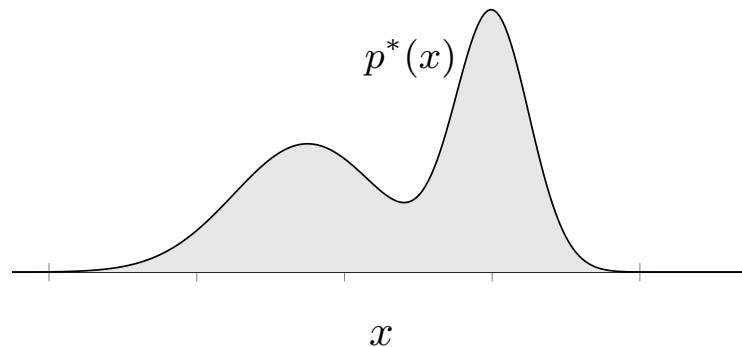
2. *Bayesian networks (20 points).* Consider six discrete random variables, A, B, C, D, E , and F . The parent set of each variable is given by:

- $\text{pa}(A) = \emptyset$,
- $\text{pa}(B) = \emptyset$,
- $\text{pa}(C) = \{A, B\}$,
- $\text{pa}(D) = \{C\}$,
- $\text{pa}(E) = \{B\}$,
- $\text{pa}(F) = \{D, E\}$.

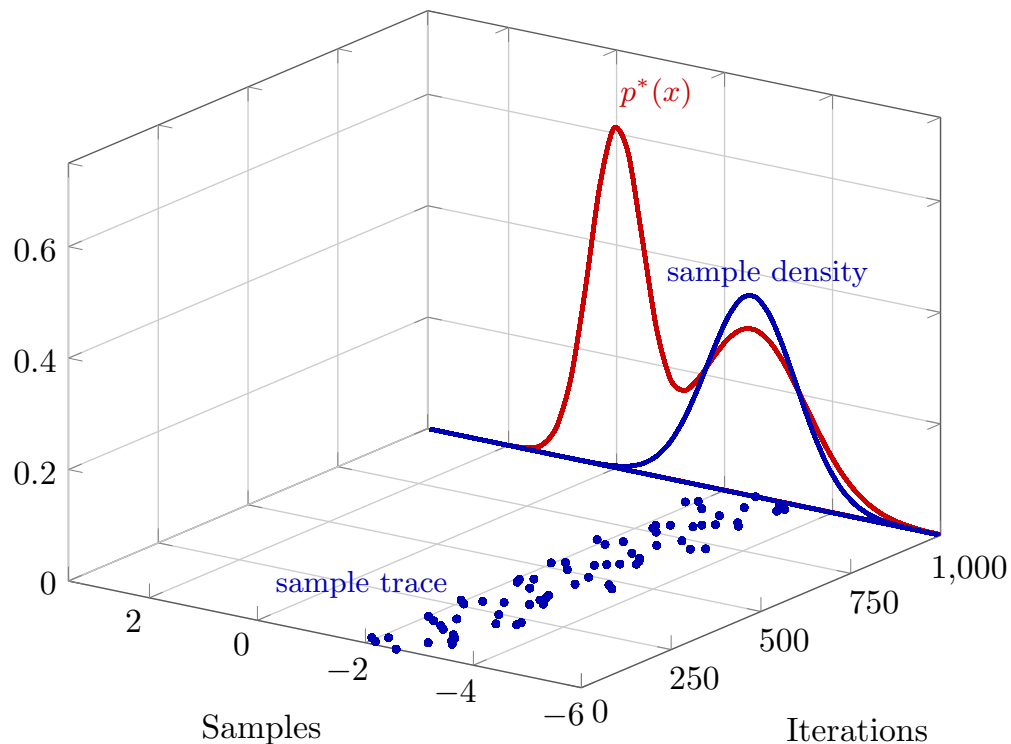
- (a) (*6 points.*) Draw the *directed acyclic graph* for this Bayesian network. Based on the structure, write down the factorization of the joint probability distribution of all variables $\mathbf{P}(A, B, C, D, E, F)$.
- (b) Using the ideas of *d-separation*, determine whether the following statements are true or false for any distribution compatible with the graph. Justify your answer by identifying the type of path, *e.g.*, chain, fork, or inverted fork (collider).
- i. (*2 points.*) $(A \perp\!\!\!\perp B)$.
 - ii. (*2 points.*) $(A \perp\!\!\!\perp B \mid D)$.
 - iii. (*2 points.*) $(D \perp\!\!\!\perp E \mid B)$.
 - iv. (*2 points.*) $(D \perp\!\!\!\perp E \mid B, F)$.
- (c) (*6 points.*) For this Bayesian network, what is the Markov blanket of each variable?

3. *Metropolis-Hastings algorithm (10 points).* Suppose we want to sample from a target distribution $p^*(x)$, which is known up to a normalizing constant, *i.e.*, $p^*(x) = \tilde{p}(x)/Z$. Let q be a proposal distribution that is easy to sample from, where $q(x' | x)$ denotes the probability of proposing x' given the current state x . We assume that q is asymmetric, *i.e.*, $q(x' | x) \neq q(x | x')$ in general.

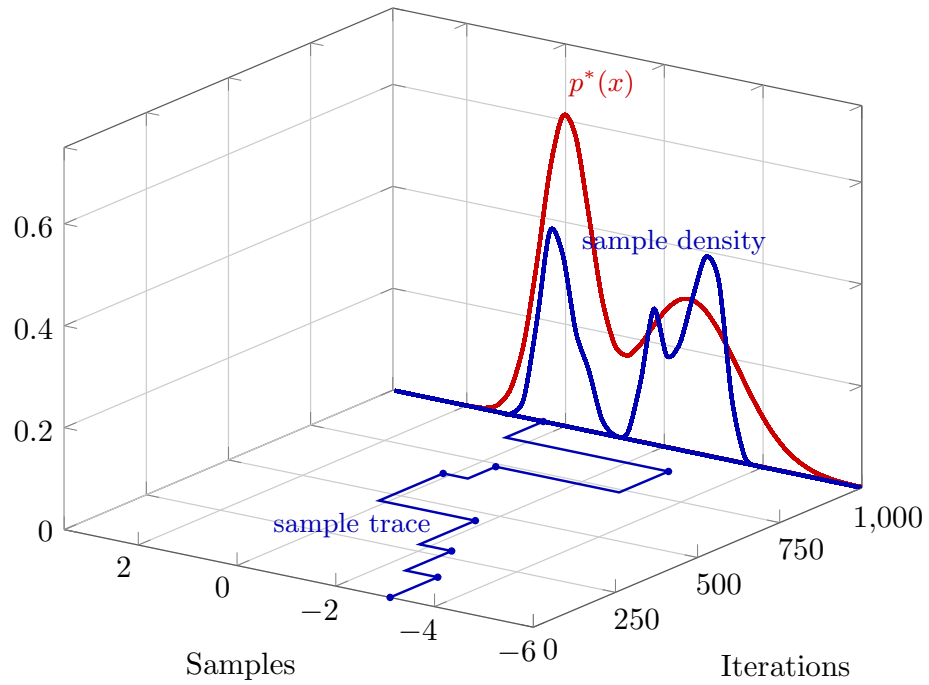
- (a) (5 points.) Describe the Metropolis-Hastings algorithm for generating samples from $p^*(x)$. Note that only the unnormalized target density $\tilde{p}(x)$ is known and can be evaluated.
- (b) Suppose the target distribution is a mixture of two Gaussian distributions shown in the figure below, and the proposal distribution is a Gaussian distribution centered at the current state x with a fixed variance σ^2 , *i.e.*, $q(x' | x) = \mathcal{N}(x, \sigma^2)$.



- i. (3 points.) Consider the case when you see the following sample distribution after running the Metropolis-Hastings algorithm for a long time.



- What is the likely reason for this sample distribution? How can you fix it?
- ii. (2 points.) Do the same reasoning for the following sample distribution.



4. *Linear output hidden Markov models (10 points).* Suppose we are given a dataset

$$(x(t), y(t)), \quad t = 1, \dots, m,$$

where $x(t) \in \mathbf{R}^n$ is the feature vector and $y(t) \in \mathbf{R}$ is the corresponding output, observed from a linear output hidden Markov model with K hidden states. It is assumed that the output $y(t)$ at each time step is generated from a linear measurement model, given by

$$y(t) = \theta_{z(t)}^T x(t) + \epsilon(t),$$

where $z(t) \in \{1, \dots, K\}$ is the hidden state at time t , $\theta_{z(t)} \in \mathbf{R}^n$ is the coefficient vector associated with state $z(t)$, and $\epsilon(t)$ is standard Gaussian noise.

We would like to estimate the parameters of the model, including the state transition probabilities P and the linear measurement coefficients $\theta_1, \dots, \theta_K$, based on the observed dataset. We consider a two-step procedure, which first estimates the coefficients θ_i together with the hidden states $z(t)$, so that in the second step the state transition probabilities P can be estimated based on the inferred hidden states.

- (a) The first step consists in minimizing the total squared error between the observed outputs and the outputs predicted by the linear output hidden Markov model. For simplicity of notation, we may transform the hidden state labels into one-hot vectors, *i.e.*, let $z(t) \in \{e_1, \dots, e_K\} \subseteq \mathbf{R}^K$, where e_i is the i -th standard basis vector in $\mathbf{R}^K$, which has 1 in the i -th entry and 0 elsewhere. Then we can formulate the first step of the procedure as an optimization problem, which is given by

$$\begin{aligned} & \text{minimize} && J(\theta_1, \dots, \theta_K, z(1), \dots, z(m)) \\ & \text{subject to} && z(t) \in \{e_1, \dots, e_K\}, \quad t = 1, \dots, m \end{aligned} \tag{1}$$

where the optimization variables are $\theta_1, \dots, \theta_K \in \mathbf{R}^n$ and $z(1), \dots, z(m) \in \mathbf{R}^K$.

- i. (*3 points.*) Please give an explicit expression of the objective J in the problem (1), as a function of the linear coefficients θ_i and the one-hot vectors $z(t)$.

In your objective, what is the role of the one-hot vector $z(t)$?

Hint. You may find it useful to write $z_i(t)$ for the i -th component of $z(t)$, so that $z_i(t) = 1$ if $z(t) = e_i$ and $z_i(t) = 0$ otherwise.

- ii. (*2 points.*) Based on your expression for J , please describe an algorithm for approximately solving the optimization problem (1). In particular, you should explain what is the objective and the variables of each subproblem in your formulation, and how they are updated.

Hint. You might get some inspiration from the *expectation-maximization* algorithm or the *k-means clustering*.

- (b) (*5 points.*) Now suppose we have obtained an estimate of the hidden states $\tilde{z}(t) \in \{1, \dots, K\}$ for all $t = 1, \dots, m$. Explain how to estimate the state transition probabilities P based on the inferred hidden states. You can either express your answer in words or mathematically.